PRODUCTION ECONOMICS FIELD STUDY

From Iron Rods to Insight

PC Jain Company: customised iron-structure manufacturing
Applying the Cobb–Douglas framework to a 26-year-old business

ABOUT THIS REPORT

This report is a field study of a real business. It uses one direct interview with the owner of PC Jain Company to describe how the business produces its output, and then uses the Cobb–Douglas production function to ask a simple question: **what limits the growth of this business, and is it getting more efficient as it grows?**

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Assignment	Individual Production Economics Assignment, Term 1
Business studied	PC Jain Company, customised iron structures for construction firms and builders
Evidence base	One structured owner interview. Figures marked (est.) are owner estimates, not audited data.

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Executive summary

PC Jain Company is a 26-year-old manufacturer of customised iron structures. Its customers are construction firms and builders. This study is based on a direct interview with the owner. It describes the inputs the business uses, how an order moves from phone call to delivery, and what that says about productivity and returns to scale.

26 years	**2 + 4 + ~11**	**~100 → ~300 m**	**~20 → ~30**
in operation	owners, permanent staff, variable workers	output per day, normal → peak (est.)	orders per week, normal → peak (est.)

Key findings

1	**Skilled labour is the binding constraint.** The owner says skilled workers are very hard to find and are a major cause of production delays. Demand and machinery are not the main limits.
2	**Capital and technology raise productivity, but they need skilled people to run them.** CNC improves efficiency, quality and accuracy, and the owner says efficiency has improved "a lot". Labour and capital behave as complements.
3	**Peak output is about three times normal output, while orders rise by about half.** This is consistent with increasing returns as labour, capital and technology grow together, but it is an indication, not proof (Section 6).
4	**Custom work creates scrap.** Every job has different specifications, so some wastage is unavoidable. No scrap percentage was collected, so the size of the problem is still unknown.

BOTTOM LINE

The business is held back by people, not machines. The recommended order of action is: **(1)** build a skilled-worker pipeline, **(2)** then expand CNC capacity so the extra machines have trained operators, and **(3)** reduce scrap with digital cutting plans.

1 Business overview

Attribute	Detail
Business	PC Jain Company: customised iron structures
Experience	26 years in operation
Customers	Construction companies and builders
People	2 owners · 4 permanent employees · ~11 variable workers
Working hours	About 9 hours per day
Orders (normal / peak)	~20 / ~30 per week (est.)
Output (normal / peak)	~100 / ~300 metres per day (est.)
Raw materials	Iron rods and steel plates
Demand driver	Demand rises when iron prices fall

How the study was done

One structured interview with the owner. The answers were grouped under labour, capital, technology and output, then mapped onto the Cobb–Douglas production function. Rough numbers are shown as estimates, and gaps are stated, not filled.

2 Inputs: labour, capital and technology

Input	What the business uses	What it means
Labour	**Permanent staff** cover estimation, customer relations, accounting and operations. **Variable workers** cover delivery and other wage-based work when needed. Core production tasks are measurement, cutting (skill-based), weighing and loading.	Lean core team that flexes with orders. Quality and production decisions depend on skilled workers.
Capital	CNC cutting, iron rod cutting machine, rust-removal equipment (new), welding and blow torches, weighing machine, transport vehicles. Some older machinery was replaced or sold (dates not provided).	A growing mechanised base. Capital adds speed and accuracy but does not replace skill.
Technology	CNC machines, Tally for accounts, phone and WhatsApp for orders and specifications.	The owner says technology has improved efficiency “a lot”. CNC improves efficiency, quality and accuracy.

What the labour evidence tells us

- Skilled workers are very difficult to find. The owner names this as the top constraint.
- Workers are also a major cause of production delays, and extra labour only raises output when there are enough orders to absorb it.

3 Production process

An order moves through nine steps from the first phone call to delivery. Measurement and cutting are the skill-based steps. They sit early in the flow, so any delay there is passed on to CNC processing, weighing, loading and delivery.

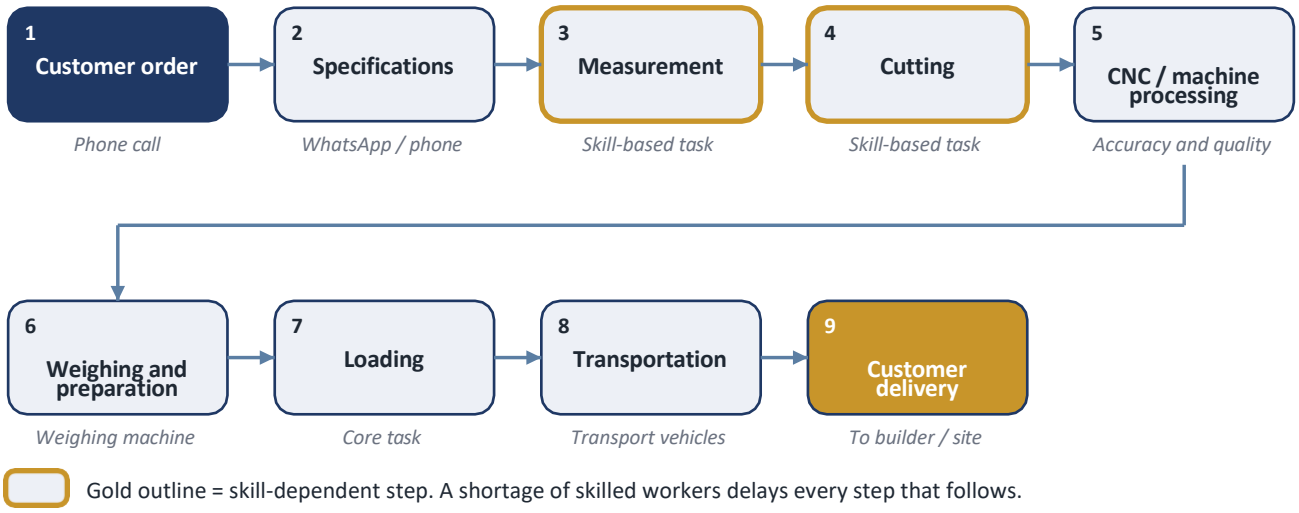


Figure 1. From customer order to delivery.

4 Productivity and wastage

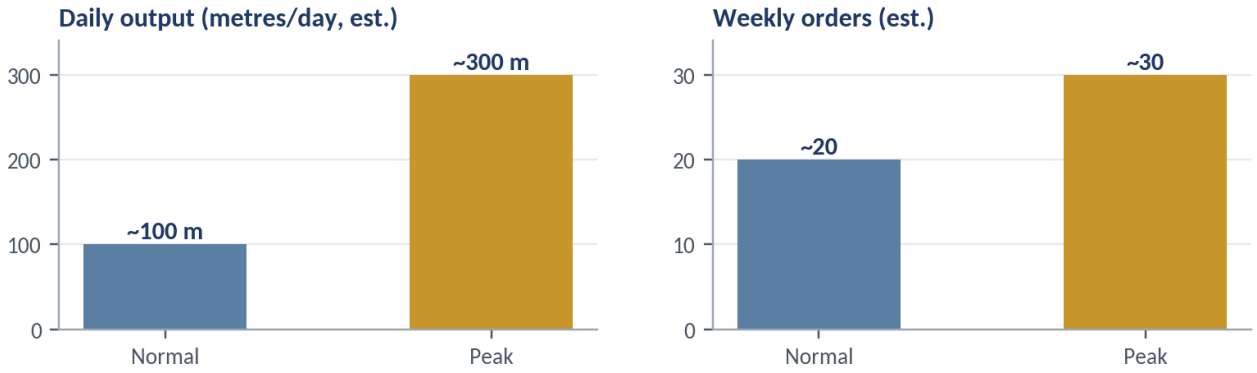


Figure 2. Normal vs peak output and orders (owner estimates).

Measure	Normal	Peak	Change
Output per day (est.)	~100 m	~300 m	~3.0×
Orders per week (est.)	~20	~30	~1.5×

- **Output grows faster than orders.** The interview did not ask why. A likely reason is that peak orders are larger or the same people and machines are used more intensively. This is an inference, not something the owner confirmed.
- **Customisation creates scrap.** Specs change from job to job, so some scrap is unavoidable. Part of it can be reduced with better cutting plans (Recommendation 3). The scrap rate was not measured.

5 Cobb–Douglas mapping

$$Q = A \cdot L^{\alpha} \cdot K^{\beta}$$

The Cobb–Douglas function says output (Q) depends on labour (L), capital (K) and a technology factor (A). It is used here as a way to organise the interview evidence. The exponents α and β were **not** estimated, because that needs input and output data over time.

Variable	Mapped to PC Jain Company
Q	Metres of structure produced: ~100 per day normal, ~300 per day peak (est.)
L	Skilled workers doing measurement, cutting, processing and loading
K	CNC, cutting machine, welding equipment, weighing machine, transport
A	CNC, computers, Tally, WhatsApp and phone coordination

What the mapping shows

- **Labour and capital are complements.** Machines lift productivity, but only skilled hands can run them. Buying more capital without skilled labour would add little.
- **Technology (A) acts as a multiplier.** CNC and digital coordination raise output from the same labour and capital by improving accuracy and speed.
- **Labour is the scarce input.** Because K and A are easier to add than skilled L, the next unit of output is limited by L.

6 Returns to scale

Returns to scale ask what happens to output when *all* inputs are increased by the same proportion. If output rises by more than that proportion, returns are increasing.

Points that support increasing returns	Gaps and caveats
Peak output is ~3× normal while orders are ~1.5× (est.).	Peak vs normal is mostly a short-run change in how hard existing capacity is used, not a true scale-up of all inputs.
Labour, capital and technology have grown together over the years (CNC, rust-removal equipment).	No before-and-after output data was collected for these purchases, and no dates were given.
The owner reports efficiency improved "a lot" with technology.	This is an opinion. Input levels at normal and peak were not measured.

CONCLUSION ON RETURNS TO SCALE

Within its current range, PC Jain Company shows **qualitative evidence consistent with increasing returns**. It is not statistical proof. The main risk is on the labour side: if skilled workers cannot be found, growth in capital and technology will stop adding output at the same rate, and returns could start to diminish.

7 Recommendations

The three recommendations are ordered by the logic of the findings. The binding constraint is skilled labour, so it comes first. Machines come second because they only pay off with trained operators.

#	Action	Evidence from the study	Expected benefit	What to watch
1	**Build internal skilled-worker training**	Skilled workers are scarce and a major constraint and cause of delays.	Removes the binding constraint and protects quality.	Time and cost of training, and keeping trained workers.
2	**Expand CNC and precision machinery**	CNC already improves efficiency, quality and accuracy.	More accurate output per skilled worker; higher peak capacity.	Only pays off if there are enough skilled operators, so sequence it with or after #1.
3	**Use digital cutting plans to cut scrap**	Custom specifications generate scrap.	Lower raw-material waste on custom jobs.	Scrap rate is not yet measured. Record a baseline first, then compare.

8 Conclusion and limitations

PC Jain Company is a lean, skill-dependent business that has built a growing base of machines and technology over 26 years. Its output can rise sharply at peak, and the evidence is consistent with increasing returns. Its limit is not demand and not machinery. It is the supply of skilled workers. Any plan for growth should start there.

Limitations

- Based on **one interview** with the owner. There was no site measurement and no second source.
- Orders and output figures are **owner estimates**.
- The Cobb–Douglas exponents were **not estimated**. The model is used as a framework, not as a fitted equation.
- No data was collected on scrap percentage, wages, machine costs or purchase dates.

What would make the findings stronger

- Labour and machine hours at normal and at peak output.
- Scrap as a percentage of material used, per job type.
- Output before and after the CNC and rust-removal purchases.

Business process infographic

LABOUR

- 2 owners, 4 permanent, ~11 variable
- ~9-hour working day
- Skilled workers are scarce
- Main production constraint

CAPITAL

- CNC cutting
- Cutting and welding equipment
- Rust-removal equipment (new)
- Weighing machine, transport

TECHNOLOGY

- CNC plus computers
- Tally for accounts
- WhatsApp / phone coordination
- Improves accuracy and efficiency

PRODUCTIVITY

- ~100 m/day normal (est.)
- ~300 m/day peak (est.)
- ~20–30 orders/week (est.)
- Custom orders create scrap

RETURNS TO SCALE

- Consistent with increasing returns
- Labour, capital, technology grow together
- Skilled labour is the risk
- Qualitative, not proven

KEY INSIGHTS

- Labour: critical
- Capital: raises productivity
- Technology: accuracy and efficiency
- Constraint: skilled workers

RECOMMENDATIONS

1 Train skilled workers internally **2** Expand CNC / precision machinery **3** Use digital cutting plans to cut scrap





Source: owner interview. (est.) = owner estimate.